

Machine Learning – NLP

Week 13 – Part 1 – Introduction to
Natural Language Processing (NLP)

CS 457 - L1 Data Science

Zeesham Rasheed

NLP involves many Data Disciplines



- Linguistics
- Cognitive Science
- Artificial Intelligence, Statistics and Machine Learning

- **Natural Language Processing is among the hottest topic in the field of data science**
 - Search Autocorrect and Autocomplete
 - Language Translator
 - Social Media Monitoring
 - Chatbots
 - Survey and Review Analysis
 - Targeted Advertising
 - Hiring and Recruitment

More on NLP Technologies



- **Speech interfaces**
 - Smart TVs, Smart Phones, Alexa, Siri, Call Centers
- **Machine translation**
 - Web content (e.g., <https://www.lemonde.fr>) to English
- **Sentiment analysis**
 - primarily social media
 - product reviews
- **Search engines**
 - auto correct and auto complete
- **Document analysis**
 - Plagiarism (SafeAssign (<https://its.gmu.edu/service/safeassign/>), TurnItIn)
 - Source identification, authorship questions

- Text mining:
 - Web pages (including scraping!)
 - Medical records
 - Customer surveys
 - Email filtering (spam)
 - DNA sequences (genes are composed of “words”... ACTC, CATG)
 - Incident reports (Police, Accident, log files)
 - Drug interaction reports (“word correlation”)
 - News stories (e.g. predict stock movements on Twitter or Reddit)

Text Processing Steps



Tokenization

- breaking text into smaller meaningful elements called tokens.
- These so-called tokens can be words, numbers, or punctuation marks.

Word Stemming and Lemmatization

- reduce an inflected (or derived) word's form to its root or base form,
- a word's endings are chopped off in hope of achieving its base form.

Part-Of-Speech (POS) Tagging

- assign labels for each token (a word in this case) with its respective grammatical component, such as noun, verb, adjective, or adverb.

Chunking

- a process that extracts phrases (sentences) from a text sample.

Stop Word Removal

- **Stop words** are simply words that have very little meaning and are mostly used as part of the grammatical structure of a sentence.
- Words like "the", "a", "an", "in", etc. are considered stop-words.

Named Entity Recognition

- Identifying entities such as *Names*, *Locations*, *Dates*, or *Organizations* that exist in an unstructured text sample.

Sentiment Analysis

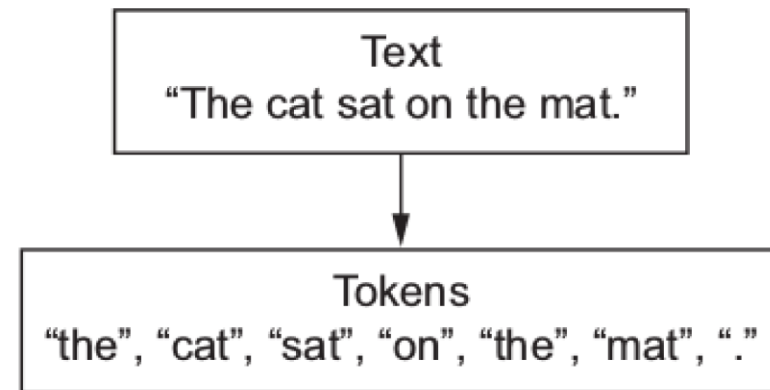
- Predicting the sentiment/mood for a given text.
- Used in analyzing customer reviews.

Text Summarization

- Summarizing text and find important keywords.
- Also called Topic Modeling

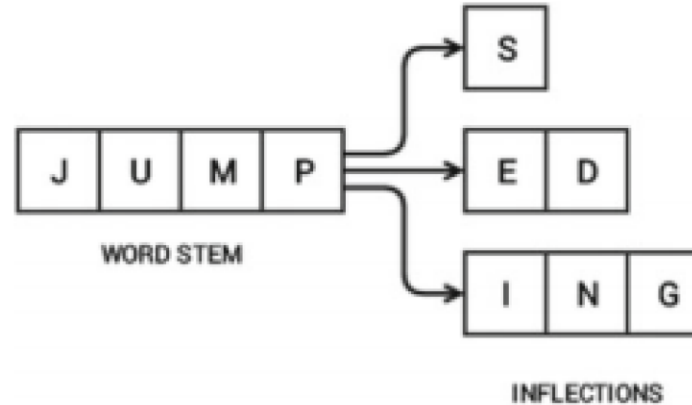
Tokenization/Segmentation

- ▶ Tokenization is the process of **breaking up** text document into **individual** words called **tokens**.
- ▶ Types of Tokenization
 - ▶ **Sentence** Tokenization
 - ▶ **Word** Tokenization



Stemming

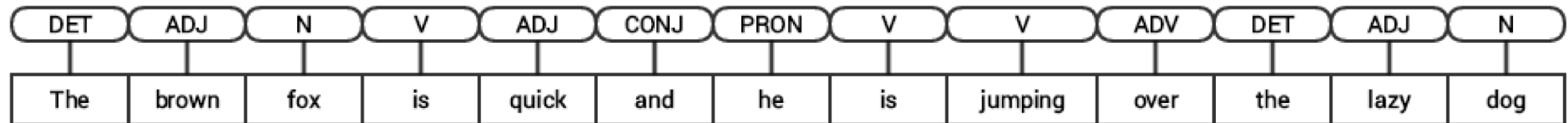
- ▶ The process of reducing a word to its **stem/root word**.
- ▶ **Reduces inflection** in words (e.g. 'help', 'helping', 'helped', 'helpful') to their root form (e.g. 'help')
- ▶ **removes** the **morphological affixes** from words, leaving only the word stem



Part-of-Speech (POS) Tagging



- Each word in a sentence can be classified into classes
- such as verbs, adjectives, nouns, etc.
- POS Tagging is a process of tagging words in a sentence to particular part of speech, based on its definition and context in the sentence.



	Word	POS tag	Tag type
0	US	NNP	PROPN
1	unveils	VBZ	VERB
2	world	NN	NOUN
3	's	POS	PART
4	most	RBS	ADV
5	powerful	JJ	ADJ
6	supercomputer	NN	NOUN
7	,	,	PUNCT
8	beats	VBZ	VERB
9	China	NNP	PROPN

SpaCy POS tagging

	Word	POS tag
0	US	NNP
1	unveils	VBZ
2	world's	VBZ
3	most	RBS
4	powerful	JJ
5	supercomputer,	JJ
6	beats	NNS
7	China	NNP

NLTK POS tagging

Natural Language Tool Kit (NLTK)

- NLTK defines an infrastructure that can be used to build NLP programs in Python.
- It provides basic classes for representing data relevant to natural language processing.
- Standard interfaces for performing tasks such as part-of-speech tagging, syntactic parsing, and text classification.
- Resources: Download at <http://www.nltk.org>

spaCy

- spaCy is a free, open-source library for advanced Natural Language Processing (NLP) in Python.
- build information extraction or natural language understanding systems, or to pre-process text for deep learning.
- Built-in models for text processing
- Standard interfaces for performing tasks such as Name Entity Recognition

Python NLP Libraries (2)



Gensim

- Gensim is a Python library for topic modelling, document indexing and similarity retrieval with large corpus
- Target audience is the natural language processing (NLP) and information retrieval (IR) community.
- Used for Topic Modeling, Keyword Extraction, Text Summarization
- Built-in Large Scale Semantic NLP Models
- <https://pypi.org/project/gensim/>

RAKE

- RAKE short for Rapid Automatic Keyword Extraction algorithm
- domain independent keyword extraction algorithm
- determine key phrases in a body of text by analyzing the frequency of word appearance and its co-occurrence with other words in the text.
- <https://pypi.org/project/rake-nltk/>
- <https://thinkinfi.com/automatic-keyword-extraction-using-rake-in-python/>

Python NLP Libraries (3)



TextBlob

- TextBlob is a Python (2 and 3) library for processing textual data.
- provides functions to common natural language processing (NLP) tasks such as part-of-speech tagging, noun phrase extraction, **sentiment analysis**, classification, translation, and more.
- <https://textblob.readthedocs.io/en/dev/>

Tweepy

- An easy-to-use Python library for accessing the Twitter API.
- <https://www.tweepy.org/>

Twitter Developer Account

- As for Twitter, REST API allows you to read and write Twitter data.
- This account gives you Consumer and Access tokens and secret keys
- <https://developer.twitter.com/en/docs/tutorials/step-by-step-guide-to-making-your-first-request-to-the-twitter-api-v2>

End of Part 1



Machine Learning – NLP

Week 13 – Part 2 – NLP Use Cases

CS 457 - L1 Data Science

Zeesham Rasheed

Language Ambiguity



- **Language is Ambiguous**
- **Especially news headlines**
 - Court to Try Shooting Defendant
 - Hospitals Sued by 7 Foot Doctors
 - FBI to Help Corrupt Police Force
 - Union Demands Increased Unemployment
 - Man Shoots Pictures of Wolf Chasing Him on Motorcycle in Canada
 - Teacher Strikes Idle Kids
 - Kids Make Nutritious Snacks
- <https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>

Part of Speech (POS) Tagging



- **NLP: Need to extract parts of speech (PoS) and syntax**
 - PoS = Noun, Verb, Object, Adjectives etc.
- Also extract/infer context and intent in a text
- Interpretation of *punctuation* and *verbal pauses* is **hard**:
 - Missing commas can be **dangerous**:
 - Have you eaten Grandma? / Have you eaten, Grandma?
 - <https://www.amazon.com/Have-You-Eaten-Grandma-Life-Saving/dp/1982127406/>
 - A panda eats, shoots, and leaves. / A panda eats shoots and leaves
 - <https://www.amazon.com/Eats-Shoots-Leaves-Tolerance-Punctuation/dp/1592402038/>
 - Aye, mate! / I'm eight!
- Ambiguities are generally resolved through context and (large) dictionary searches

Name Entity Recognition (NER)



- **Named Entity recognition (NER)**

- Probably the first step towards information extraction that seeks to locate and classify named entities in text into pre-defined categories
 - such as the names of persons, organizations, locations, expressions of times, quantities, monetary values, percentages, etc.
- NER is used in many fields in Natural Language Processing (NLP), and it can help answering many real-world questions, such as:
 - Which companies were mentioned in the news article?
 - Were specified products mentioned in complaints or reviews?
 - Does the tweet contain the name of a person? Does the tweet contain this person's location?

- **How to build Named Entity Recognizer (NER)?**
- With **NLTK** and **SpaCy** libraries in Python to identify the names of things, such as persons, organizations, or locations in the raw text.
- **SpaCy Model for Entity Recognition**
 - Built on Convolutional Neural Networks (CNN) Deep Learning algorithm
 - Assigns context-specific token vectors, POS tags, dependency parse and named entities.
 - <https://spacy.io/models/en>
 - English multi-task CNN trained on OntoNotes (Linguistic Dataset).

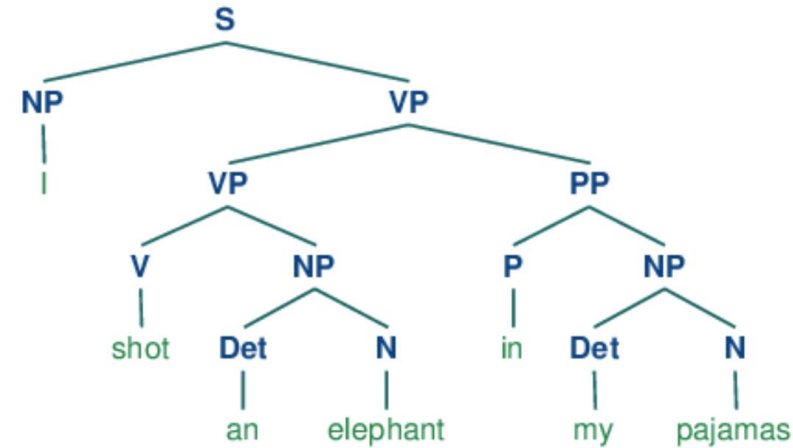
- **About OntoNotes**

- OntoNotes Release 5.0 is the final release of the OntoNotes project, a collaborative effort between BBN Technologies, the University of Colorado, the University of Pennsylvania and the University of Southern Californias Information Sciences Institute.
- The goal of the project was to annotate a large corpus comprising various genres of text (news, conversational telephone speech, weblogs, newsgroups, broadcast, talk shows) in three languages (English, Chinese, and Arabic) with structural information (syntax and predicate argument structure) and shallow semantics (word sense linked to an ontology and coreference).
- <https://catalog.ldc.upenn.edu/LDC2013T19>

Language Ambiguity Example



- **Sample Text:**
- *"I shot an elephant in my pajamas"*
- ambiguity in the phrase can be analyzed using PoS tree



- **Part of Speech Tagging**
- CARDINAL, DATE, EVENT,
- GPE, LANGUAGE, ORG,
- PERSON, PRODUCT,
- QUANTITY, TIME etc.

inspector general, **Michael E. Horowitz** **PERSON**, that preceded Mr. **Strzok** **PERSON**'s firing not only criticized his conduct in sending the texts but also his use of personal email accounts to handle sensitive information., In addition, the inspector general criticized Mr. **Strzok** **PERSON**'s decision not to move swiftly to examine new emails related to the **Clinton** **PERSON** investigation **just weeks** **DATE** before the **2016** **DATE** election., Mr. **Horowitz** **PERSON** said in his report that he was "deeply troubled" by the text messages., **Hundreds** **CARDINAL** exchanged **over months** **DATE** were found in which the pair disparaged Mr. **Trump** **PERSON** and, to a lesser extent, Mrs. **Clinton** **PERSON**, exchanged work gossip and bantered., On **Twitter** **PRODUCT**, Mr. **Strzok** **PERSON** said he was "deeply saddened by this decision," adding, "It has been an honor to serve my country and work with the fine men and women of the FBI."Mr., **Strzok** **PERSON** became emblematic of Mr. **Trump** **PERSON**'s unfounded assertions that a so-called deep state of bureaucrats opposed to him was undermining his presidency., Mr. **Trump** **PERSON** contended that Mr. **Strzok** **PERSON** targeted the president and accused Mr. **Strzok** **PERSON** of being "treasonous" and a "disgrace.", Mr. **Strzok** **PERSON** told lawmakers that he never leaked information about the **Russia** **GPE** inquiry, which could have upended the election and hurt Mr. **Trump** **PERSON**'s chances of becoming president., After Mr. **Horowitz** **PERSON** uncovered the texts, Mr. **Mueller** **PERSON**, who had by then taken over the investigation, removed Mr. **Strzok** **PERSON** from his team., He was reassigned to the **F.B.I.'s** **ORG** human resources division., Ms. **Page** **PERSON**, who had left Mr. **Mueller** **PERSON**'s team before the discovery of the text messages,

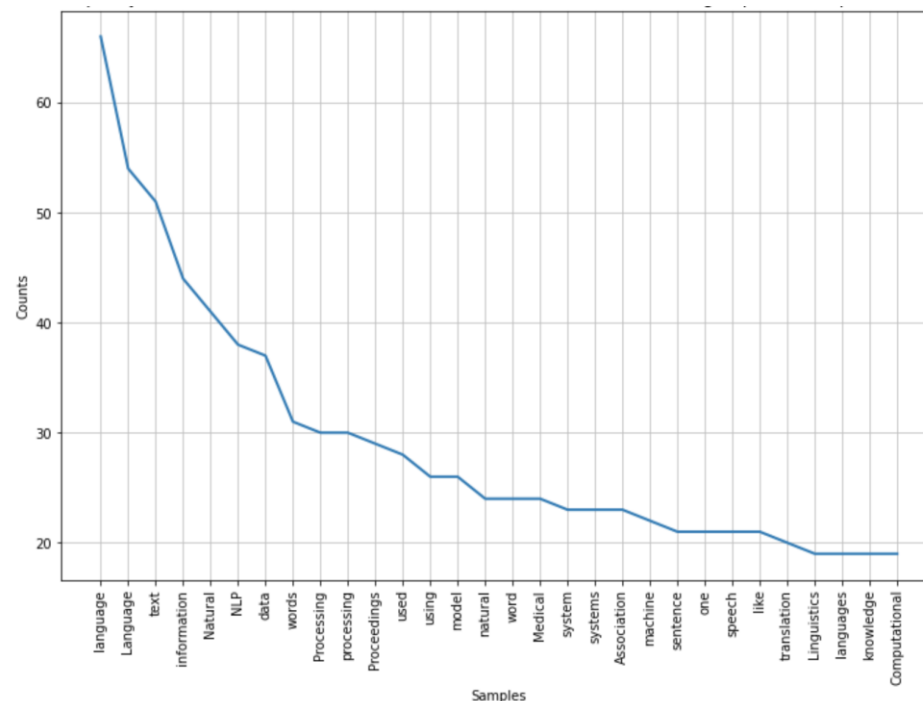
Keyword Extraction and Summarization Example



- Give any text, URL or document
- We can perform
 - **Word Frequency**
 - **Word Cloud**
 - **Extracting Keywords**

```
[31]: keywords
```

```
[31]: ['languages',  
      'texting',  
      'data',  
      'modeling',  
      'likely',  
      'word',  
      'linguist',  
      'process',  
      'information',  
      'document']
```



Why Sentiment Analysis?



- **Movie:** is this review positive or negative?
- **Products:** what do people think about the new iPhone?
- **Public sentiment:** how is consumer confidence? Is despair increasing?
- **Politics:** what do people think about this candidate or issue?
- **Prediction:** predict election outcomes or market trends from sentiment

Positive or negative movie review?

-  unbelievably disappointing
-  Full of zany characters and richly applied satire, and some great plot twists
-  this is the greatest screwball comedy ever filmed
-  It was pathetic. The worst part about it was the boxing scenes.



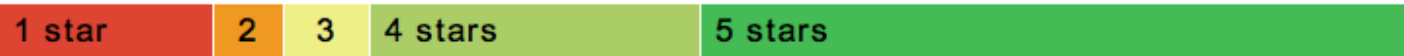
HP Officejet 6500A Plus e-All-in-One Color Ink-jet - Fax / copier / printer / scanner

\$89 online, \$100 nearby ★★★★★ 377 reviews

September 2010 - Printer - HP - Inkjet - Office - Copier - Color - Scanner - Fax - 250 sh

Reviews

Summary - Based on 377 reviews



What people are saying

ease of use		"This was very easy to setup to four computers."
value		"Appreciate good quality at a fair price."
setup		"Overall pretty easy setup."
customer service		"I DO like honest tech support people."
size		"Pretty Paper weight."
mode		"Photos were fair on the high quality mode."
colors		"Full color prints came out with great quality."

HP Officejet 6500A E710N Multifunction Printer

[Product summary](#) [Find best price](#) **Customer reviews** [Specifications](#) [Related items](#)



\$121.53 - \$242.39 (14 stores)

☐ Compare

Average rating ★★★★★★ (144)



Most mentioned



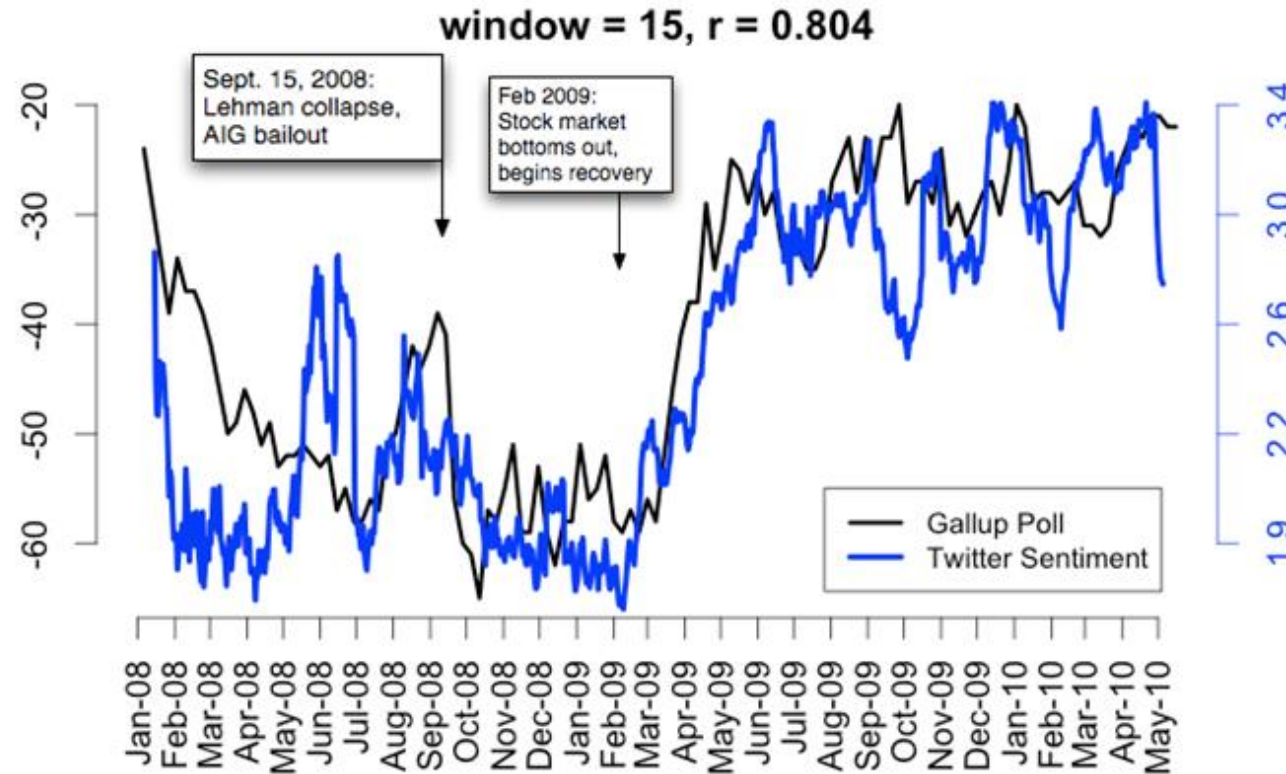
Show reviews by source

[Best Buy \(140\)](#)
[CNET \(5\)](#)
[Amazon.com \(3\)](#)

Power of Sentiment Analysis



- **Twitter sentiment versus GallupPoll of Consumer Confidence**



Brendan O'Connor, Ramnath Balasubramanyan, Bryan R. Routledge, and Noah A. Smith. 2010. From Tweets to Polls: Linking Text Sentiment to Public Opinion Time Series. In ICWSM-2010

- Johan Bollen, Huina Mao, Xiaojun Zeng. 2011.
 - Journal of Computational Science 2:1, 1-8. 10.1016/j.jocs.2010.12.007.
 - Twitter mood predicts the stock market

Twitter Based Sentiment Analysis App

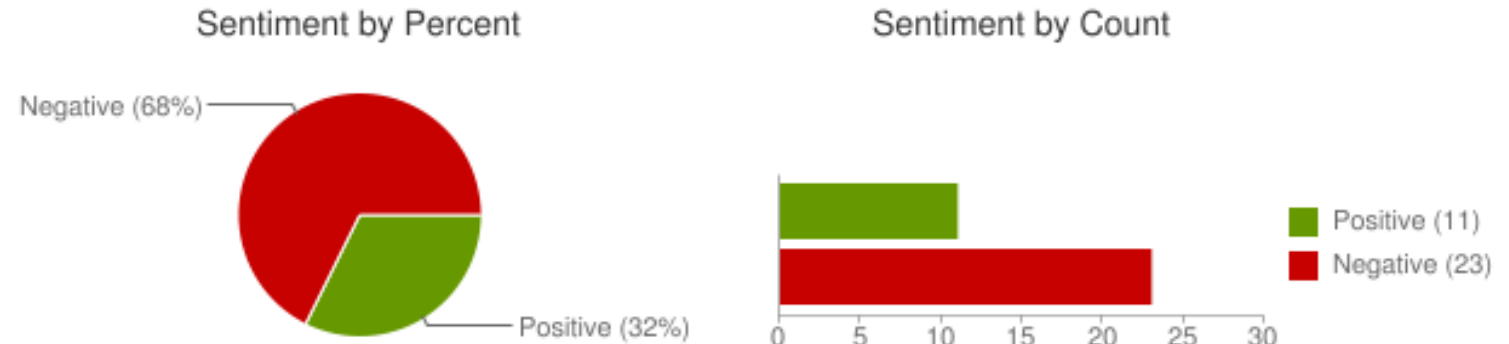


- Alec Go, Richa Bhayani, Lei Huang. 2009.
- Twitter Sentiment Classification using Distant Supervision

Type in a word and we'll highlight the good and the bad

[Save this search](#)

Sentiment analysis for "united airlines"



jjacobson: OMG... Could @United airlines have worse customer service? W8g now 15 minutes on hold 4 questions about a flight 2DAY that need a human.
Posted 2 hours ago

12345clumsy6789: I hate United Airlines Ceiling!!! Fukn impossible to get my conduit in this damn mess! ?
Posted 2 hours ago

EMLandPRGbelgiu: EML/PRG fly with Q8 united airlines and 24seven to an exotic destination. <http://t.co/Z9QloAjF>
Posted 2 hours ago

CountAdam: FANTASTIC customer service from United Airlines at XNA today. Is tweet more, but cell phones off now!
Posted 4 hours ago

- Opinion extraction
- Opinion mining
- Sentiment mining
- Subjectivity analysis

- Simplest task:
 - Is the attitude of this text positive or negative?
- More complex:
 - Rank the attitude of this text from 1 to 5
- Advanced:
 - Detect the target, source, or complex attitude types

Scherer Typology of Affective States (1984)



- **Emotion:** brief organically synchronized ... evaluation of a major event
 - *angry, sad, joyful, fearful, ashamed, proud, elated*
- **Mood:** diffuse non-caused low-intensity long-duration change in subjective feeling
 - *cheerful, gloomy, irritable, listless, depressed, buoyant*
- **Interpersonal stances:** affective stance toward another person in a specific interaction
 - *friendly, flirtatious, distant, cold, warm, supportive, contemptuous*
- **Attitudes:** enduring, affectively colored beliefs, dispositions towards objects or persons
 - *liking, loving, hating, valuing, desiring*
- **Personality traits:** stable personality dispositions and typical behavior tendencies
 - *nervous, anxious, reckless, morose, hostile, jealous*

- Polarity detection:
 - Is an IMDB movie review positive or negative?
- **Data: *Polarity Data 2.0*:**
 - <http://www.cs.cornell.edu/people/pabo/movie-review-data>

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan. 2002. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP-2002, 79—86.

Bo Pang and Lillian Lee. 2004. A Sentimental Education: Sentiment Analysis Using Subjectivity Summarization Based on Minimum Cuts. ACL, 271-278

Sentiment Analysis using TextBlob



- **TextBlob Python Library**
- The **textblob.sentiments** module contains two sentiment analysis implementations, PatternAnalyzer (based on the pattern library) and NaiveBayesAnalyzer (an NLTK classifier trained on a movie reviews corpus).
- You can also create your own classifiers with different labels.
- <https://textblob.readthedocs.io/en/dev/classifiers.html#loading-data-and-creating-a-classifier>

Creating Your Own Classifier



Loading Data and Creating a Classifier

First we'll create some training and test data.

```
>>> train = [  
...     ('I love this sandwich.', 'pos'),  
...     ('this is an amazing place!', 'pos'),  
...     ('I feel very good about these beers.', 'pos'),  
...     ('this is my best work.', 'pos'),  
...     ("what an awesome view", 'pos'),  
...     ('I do not like this restaurant', 'neg'),  
...     ('I am tired of this stuff.', 'neg'),  
...     ("I can't deal with this", 'neg'),  
...     ('he is my sworn enemy!', 'neg'),  
...     ('my boss is horrible.', 'neg')  
... ]  
>>> test = [  
...     ('the beer was good.', 'pos'),  
...     ('I do not enjoy my job', 'neg'),  
...     ("I ain't feeling dandy today.", 'neg'),  
...     ("I feel amazing!", 'pos'),  
...     ('Gary is a friend of mine.', 'pos'),  
...     ("I can't believe I'm doing this.", 'neg')  
... ]
```

Now we'll create a Naive Bayes classifier, passing the training data into the constructor.

```
>>> from textblob.classifiers import NaiveBayesClassifier  
>>> cl = NaiveBayesClassifier(train)
```

TextBlob – Python Example



```
from textblob import TextBlob
```

```
[56]: s = TextBlob('AIT 580 is the best course throughout my course work')  
s.sentiment
```

```
[56]: Sentiment(polarity=1.0, subjectivity=0.3)
```

```
[57]: s = TextBlob('AIT 580 is an okay course')  
s.sentiment
```

```
[57]: Sentiment(polarity=0.5, subjectivity=0.5)
```

```
[55]: s = TextBlob('AIT 580 is the worst course I have ever taken')  
s.sentiment
```

```
[55]: Sentiment(polarity=-1.0, subjectivity=1.0)
```

Some Interesting Demo and Applications



- Sentiment Analysis with Python NLTK Text Classification
 - <http://text-processing.com/demo/sentiment/>
- Tweet Sentiment Visualization Tool
 - https://www.csc2.ncsu.edu/faculty/healey/tweet_viz/tweet_app/
- One Million Tweet Map
 - <https://onemilliontweetmap.com>
- Analyzing Messy Data Sentiment with Python and nltk
 - <https://www.twilio.com/blog/2017/09/sentiment-analysis-python-messy-data-nltk.html>
- DSC Webinar Series: The Promise of Natural Language Processing (NLP)
 - <https://www.datasciencecentral.com/video/dsc-webinar-series-the-promise-of-natural-language-processing-nlp>

End of Part 2

